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Digital transformation and hospitality management competencies: Toward an integrative framework

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ABSTRACT

Digital transformation requires hospitality and tourism organizations to build and sustain digital business capabilities such as digital customer engagement, digital customer experience management, digital innovation, digital leadership, and others. These organization digital business capabilities, in turn, require corresponding digital transformation and digital business management competencies. This has important implications for required hospitality management competencies. We identify the specific digital transformation and digital business management competencies required by hospitality managers. Then, given a disconnect between these competencies and previously identified hospitality management digital technology competencies, we propose a framework integrating digital transformation and digital business management competencies with previously identified digital technology competencies required by hospitality managers. The proposed framework has important implications for curricula design and managerial practice. These implications as well as future research ideas are discussed.

1. Introduction

Digital technologies (DTs) are combinations of information, computing, communication, and connectivity technologies that include social, mobile, analytics, cloud computing, internet of things, artificial intelligence, and blockchain technologies (e.g., see Vial, 2019; Bharadwaj et al., 2013). As DTs have advanced exponentially, the imperative for all organizations, including hospitality and tourism (H&T) organizations, to adapt to the resultant changes has evolved from digitization, to digitalization, to digital transformation (e.g., see Lam and Law, 2019; Vial, 2019; Hess et al., 2016). Digitization required converting information into digital format so that it could be stored, processed, and or transmitted (e.g., Gartner, n.d.-a; Ritter and Pedersen, 2020). In contrast, digitalization required H&T organizations to use DTs and digitized data to transform roles, processes, and workflows so as to simplify established ways of working and improve efficiency (e.g., see Gartner, n.d.-b; Ritter and Pedersen, 2020). Building on this, the digital transformation or digital business transformation imperative now requires H&T organizations to re-examine their businesses from end to end, and to redesign them for game changing efficiencies, breakthroughs in customer experience and value, scalability, optimization of strategic agility, and minimization of strategic risk (e.g., see Lam and Law, 2019; Buhalis et al., 2019; Vial, 2019; Hess et al., 2016; Gartner, n.d.-c). Effectively delivering on this digital transformation imperative requires new or enhanced organization capabilities such as digital innovation (e.g., see Warner and Wäger, 2019; Nambisan et al., 2017), digital customer engagement (e.g., see Eigenraam et al., 2018), digital customer experience management (e.g., see Weill and Woerner, 2013), and more. As used in this study, a capability refers to the organization routines/activities, structures, and or processes that come together to enable an organization to repeatably and reliably do particular things, or to achieve particular outcomes (e.g., see O'Reilly III and Tushman, 2008; Helfat and Winter, 2011). And in referring to the H&T industry, H&T organizations, or hospitality management, we mean each in the broadest sense and accepting their cited diversity and complexity (e.g., see Ottenbacher et al., 2009). That is, we include organizations and managers in subsets of the industry such as travel services, lodging services, food services, and other tourism or travel related services (e.g., see Slattery, 2002; Ottenbacher et al., 2009). We interchangeably refer to these industry subsets as sectors or segments, notwithstanding that the exact boundaries between industry sectors and segments may not be so clear cut (e.g., see Ottenbacher et al., 2009).

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The new or enhanced capabilities required by H&T organizations to respond to the digital transformation imperative have important implications for required hospitality management competencies. For example, since hospitality managers must play crucial leadership and management roles in change efforts to build and sustain these capabilities, it follows that they need corresponding new or enhanced competencies. In referring to competencies, we mean the knowledge, skills, abilities, and other characteristics that managers require to effectively do their managerial work (e.g., see Sisson and Adams, 2013; Oberländer et al., 2020). The hospitality management research has long been concerned with keeping future workforce competencies relevant to changing industry competency requirements (e.g., Buergermeister, 1983; Tas, 1988; Umbreit, 1992; Kay and Russette, 2000; Weber et al., 2013). And the subset of this research exploring required DT competencies has already identified many required competencies (e.g., Breiter and Hoart, 2000; Dopson, 2005; Bilgihan et al., 2014). But, perhaps due to the exponential pace and breadth of DT advancements (e.g., see Doorsamy et al., 2020; Steinicke, 2016), there now seems to be a disconnect between the competencies thus far identified in this literature, the organization capabilities required for digital transformation, and the resultant management competencies required to build and sustain these capabilities. An implication of this disconnect is that existing hospitality management DT competencies lists may need extension.

In addition to the disconnect, the collective list of existing DT competencies is unwieldy – encompassing nearly 100 competencies. While scholars have begun to group these competencies into meaningful categories, a lot of ambiguity remains due to overlapping and conflicting competency categories, as well as opaqueness regarding the meaning and interrelationships between particular competency categories and specific competencies. Further, the literature is not clear on which competencies are of priority, and in which situations they are of priority. The latter is of significance, since scholars have pointed to potential differences in DT competency requirements across lower level and more experienced managers (e.g., Kay and Moncarz, 2004; Kay and Russette, 2000), across types of hospitality management roles (e.g., Priyangika et al., 2020), and across H&T segments (e.g., see Breiter and Hoart, 2000). Taken together, these unresolved issues suggest a lack of specific understanding about what DT competencies hospitality managers require to effectively participate in digital transformation. These unresolved issues also suggest a lack of understanding about the inter-relationships between such required competencies and the DT competencies previously identified in the literature - as well as about the fit and interrelationships of the combined DT competency requirements with broader hospitality management competency requirements. Left unaddressed, these gaps in understanding are likely to hinder progress of this critical literature. In turn, this is likely to hinder H&T management education institutions' ability to prepare the future management workforce for effectively managing in digitally transforming or digital businesses.

A potential flow on effect of these gaps in understanding is an industry ill-equipped to effectively adapt to digital disruption – digital disruption refers to DT induced changes that prevent organizations from continuing to operate as they have always operated (e.g., see Vial, 2019). The H&T industry has gone through significant digital disruption and continues to experience it at an exponential rate (e.g., Buhalis et al., 2019). See Table 2 for specific examples of manifestations of digital disruption within the industry. A consequence of not effectively adapting to digital disruption is not being able to seize the game changing opportunities offered by DT advancements (we discuss these opportunities in more detail later, and also provide specific examples of their applications within the industry in Table 2). Given the fundamental importance to the industry of resolving these gaps in understanding, addressing them is the focus of this study. Specifically, we address the gaps by first examining and, where necessary, expanding the required managerial DT competencies to reflect digital transformation and digital business-related managerial DT competency requirements. Subsequently, we reconceptualize the expanded competencies list into an integrative framework that better clarifies how the different competencies interrelate and interact with each other. The framework also explains how DT competencies fit with broader hospitality management competencies to shape preparedness of the future hospitality management workforce to effectively participate in digital transformation and digital business change agendas at their organizations.

We conducted this research using conceptual research methods, which are suitable for research such as ours which has established domain theory and existing frameworks or conceptualizations of the phenomena - but where new developments render those conceptualizations or frameworks incomplete. As a result, their predictive and explanatory power is hampered (e.g., see Jaakkola, 2020). Through research designs that draw on theory synthesis, theory adaptation, typology, and modeling, conceptual methods enable taking stock of new developments, identification of new domain theory constructs and relationships, and therefore extension or refinement of existing conceptualizations (e.g., see Jaakkola, 2020; Lukka and Vinnari, 2014; Meredith, 1993). In doing so, conceptual methods make it easier for studies such as ours to integrate fragmented knowledge, and to provide complete and more specific overall understanding of focal phenomena. Methodologically, conceptual methods have specific requirements relating to the choice of theory used, how that theory is used, how any claims or arguments are substantiated, and what assumptions can underpin claims or arguments (e.g., see Jaakkola, 2020; Hirschheim, 2008). Adhering to these requirements, this study aims to first take stock of previously identified hospitality management DT competencies, and to identify the additional required digital transformation and digital business-related management competencies. Then, an integrative framework will be proposed that integrates the existing and additional DT competencies to provide more specific and complete overall understanding of required hospitality management DT competencies.

2. Background and literature review

2.1. Hospitality management competencies

Hospitality management researchers have investigated future management workforce competency requirements from as far back as the early 1920s (e.g., see Buergermeister, 1983; Tas, 1988; Goodman Jr and Sprague, 1991; Umbreit, 1992; Kay and Russette, 2000; Weber et al., 2013). To this end, researchers have investigated future workforce competency requirements from the perspective of H&T hiring managers (e.g., Tas, 1988; Christou, 2002; Raybould and Wilkins, 2005; Watson, 2008), from the perspective of H&T management students (e.g., Lee et al., 2010; Lee, Huh and Jones, 2016), and from the perspective of H&T management academics (e.g., Huang and Lin, 2010). Several researchers have also focused their investigations on whether competency requirements differ in specific segments of the H&T industry (e.g., Tas, LaBrecque and Clayton, 1996; Perdue et al., 2001; Kay and Moncarz, 2004), whether they differ in specific geographies (e.g., Yang et al., 2014; Predvoditeleva, Reshetnikova and Slevitch, 2019), and whether they differ across particular parts of the H&T value chain (e.g., Cheng and Bosselman, 2016). Perhaps due to these diverse focuses, scholars have identified and discussed a multitude of required competencies. These competencies have evolved over time (e.g., see Buergermeister, 1983; Tas, 1988; Umbreit, 1992; Nelson and Dopson, 2001; Raybould and Wilkins, 2005; Cheung et al., 2010; Yang et al., 2014). As they evolved, and the number of required competencies grew, researchers sought to categorize them into meaningful groups of related competencies, in order to better guide education and practice. Examples of these competency categories include Sandwith's (1993) managerial competency domains model, Chung's (2000) competency categories for hotel employees, and the Chung-Herrera et al. (2003) leadership competency model. Other examples include Tesone and Ricci's (2005) practitioner competency expectations of hospitality management graduates, and Sisson and Adams' (2013) discussion of various scholars dividing competencies into hard and soft competency categories.

2.2. DT Competencies as a subset of hospitality management competencies

In most early hospitality management competency requirements studies, DT competencies were either overlooked and not included in required hospitality management competency lists, or they were included but not considered so important (e.g., see Johanson et al., 2011). Johanson et al. (2011) note that where such competencies were discussed, the focus was largely on skills such as word processing, spreadsheet, and presentation software. But over time, scholars began to consider the significance, implications, and benefits of different DT advancements for the industry (e.g., Poon, 1993; Zabel, 1997; Buhalis, 2003; Buhalis and O'Connor, 2005). Perhaps aided by insights from some of these investigations, scholars began to explore the DT competencies required by hospitality managers (e.g., Zabel, 1997; Breiter and Hoart, 2000; Mandabach et al., 2001; Perdue et al., 2001; Yang et al., 2014). In reviewing the literature discussing required DT competencies, we found that scholars varied in the terminology they used to describe DT competencies as a group (see column two of Table 1 for terms used to refer to DT competencies over time). Notwithstanding differences in reference terms used, scholars began to identify a wide-ranging number of specific DT competencies required by hospitality managers (see full list of these competencies in Table 6 – only those with a citation). Like the evolution of broad hospitality management competencies discussed earlier, as the number of required DT competencies grew, scholars sought to organize them into meaningful categories to guide education and practice (see column three of Table 1).

Table 1
DT competency categories identified in the hospitality management literature.

Authors	Term used to refer to DT competencies	DT competency categories identified
Su et al. (1997)	Management information systems	• Management information systems
Cho and Connolly (1999) cited in Sisson and Adams (2013)	IT competencies	• Proficiency in use of applications • Behavioral and analytical abilities
Breiter and Hoart (2000)	IT competencies	N/A
Mandabach et al. (2001)	Technology skills	• General software applications • Industry software applications
Gursoy and Swanger (2004)	Computer/information technology	N/A
Kay and Moncarz (2004)	IT KSAs (knowledge, skills, abilities)	
Lowry and Flohr (2005)	Technological skills	
Dopson (2005)	E-commerce competencies	• Technology analysis and application • Information architecture management and design • Transaction related activities • Website enhancement • General use of technology
Raybould and Wilkins (2005)	Information technology usage	N/A
Johanson et al. (2011)	Computer related skills	N/A
Sisson & Adams (2013)	Using computers effectively	N/A
Bilgihan et al. (2014)	Information technology skills	• End-user applications • Industry specific applications • Conceptual/strategically focused applications

2.3. Digital disruption and digital transformation of H&T organizations – implications for hospitality management DT competencies

Hospitality management, information systems, and management researchers have established that ongoing DT advancements are inducing digital disruption (e.g., Buhalis et al., 2019; Vial, 2019). At an organization level, this digital disruption manifests itself as disruptions of customer expectations and behaviors, disruptions of the competitive landscape and bases of competition, and disruptions of data availability (Vial, 2019). In relation to disruption of customer expectations and behaviors, customers are increasingly aware of the breakthroughs in simplicity, convenience, and value possible from DT advancements. As a result, they expect the products/services they buy to reflect this simplicity, convenience, and value. Subsequently, they alter their behaviors or routines to benefit from this simplicity, convenience, and value. In relation to disruption of the competitive landscape and bases of competition, the introduction of new DTs or improvements in existing ones can lower barriers to entry into the H&T industry, allowing new start-ups or existing organizations in different industries to enter the industry (e.g., see Vial, 2019). And, in relation to disruption of data availability, both competitors and consumers have access to an unprecedented and growing amount of data. Consumers can use this data to better evaluate product/service offerings. And competitors can weaponize this data through sophisticated analytics for new or enhanced competitive advantages (e.g., superior customer attraction, superior customer experience, superior product innovation). Table 2 provides examples of manifestations of digital disruption within the H&T industry. These digital disruptions induced by DT advancements create existential threats for H&T organizations (e.g., see Buhalis et al., 2019; Vial, 2019; Sebastian et al., 2017). That is, if H&T organizations continue to operate as they have always done and do not adapt to the changed customer expectations and behaviors, to the changed competitive landscapes and bases of competition, and to the changed availability of data, they risk being replaced by competitors who do so.

But as well as creating existential threats for H&T organizations, digital disruption also brings about game changing opportunities that H&T organizations can seize (e.g., see Perelygina et al., 2021; Buhalis et al., 2019; Vial, 2019; Sebastian et al., 2017). Examples of such opportunities include offering significantly enhanced or even new disruptive products and services (e.g., Kuo et al., 2017; Filimonau and Naumova, 2020; Alrawadieh et al., 2021), bypassing intermediaries to interact directly with consumers or other stakeholders, and enhancing collaboration and coordination among strategic partners in value chains (e.g., Buhalis and Leung, 2018; Oskam and Boswijk, 2016; Perelygina et al., 2021). Additional examples of these opportunities include creating and managing ecosystem-based business models like AirBnB's (e.g., Perelygina et al., 2021; Buhalis and Leung, 2018; Oskam and Boswijk, 2016), and significantly enhancing the adaptability, agility, and ambidexterity of H&T organizations (e.g., Kuo et al., 2017; Stylos et al., 2021; Buhalis and Leung, 2018; Vial, 2019; Warner and Wäger, 2019). Examples of manifestations of the seizure of these opportunities within H&T are shown in Table 2. For H&T organizations to adapt to digital disruption, and to seize the game changing opportunities offered by DT advancements, information systems and management researchers have established that they need to become and effectively compete as digital businesses (e.g., see Vial, 2019). A digital business is a business that leverages DTs to create, facilitate, or significantly enhance business models, business processes, product and service offerings, assets, competitive advantages, and business longevity (e.g., see Warner and Wäger, 2019; Bharadwaj et al., 2013; Perelygina et al., 2021). For H&T organizations to become digital businesses, they must undertake digital transformation, which is defined as the change process involved in becoming a digital business (e.g., Lam and Law, 2019; Farías and Cancino, 2021; Vial, 2019; Warner and Wäger, 2019; Matt et al., 2015). This

Table 2
Example manifestations of digital disruption induced existential threats and game changing opportunities within the H&T industry (adapted from Busulwa et al., 2020).

Existential threat or game changing opportunity	Type of digital disruption or game changing opportunity	Examples of manifestations within the H&T industry
Existential Threat	Disruptions of customer expectations and behaviors	• Growing expectations of high personalization, serendipitous experiences, hospitableness, and a sense of community - which competitors like AirBnB are enabling through technology • Customers are increasingly basing purchasing decisions on online reviews or social media information • Customers are increasingly purchasing through aggregation platforms like Kayak, TripAdvisor, and HotelsCombined
	Disruptions of the competitive landscape and bases of competition	• Introduction and growth of home-sharing platforms (e.g., AirBnB, Vrbo, FlipKey) • Introduction and growth of online travel agencies (e.g., Trivago, Expedia) • Erosion of control over bases of competition such as setting service standards and prices, managing the end – to – end customer experience, shaping perceived customer value, and shaping brand image • Emergence of bases of competition such as data collection/integration/analytics, platform and workflow integration/automation, digital transformation/digital business strategy
	Disruptions of data availability	• Exponential growth in data volume, variety, velocity, veracity applicable to the H&T industry • Entry of big data platforms or service providers into H&T industry (e.g., Duetto) • Expanding availability and comparability of online H&T review data (e.g., TripAdvisor, Google reviews, Facebook) • Availability of sophisticated H&T data analytics tools for customer targeting, customer experience optimization, branding, and pricing/revenue optimization
Game Changing Opportunity	Offering enhanced or even new disruptive products and services	• Atlantis Dubai's virtual travel tours, Amadeus' virtual booking processes, The Hub Hotel's interactive hotel rooms, Pokémon Go–style gamified hotel/tourism entertainment, Best Western Kelowna's Augmented Reality Quest • Hilton's robot concierge "Connie", TravelMate Robotics' robot suitcase, Timbre and Yotel's drone waiters • ChefJet and Foodini's 3D printed culinary designs • Using blockchain platforms like Trippki for managing guest identification and loyalty
	Bypassing intermediaries to interact directly with consumers or other stakeholders	• AirBnB bypassing OTAs and other intermediaries • Nordic Choice Hotels and Etihad partnering with Winding Tree to bypass third parties and distribute products and services directly by leveraging blockchain technology
	Enhancing collaboration and coordination among strategic partners	• Accor's use of their MAX and WeMAX ecosystem to enhance collaboration among and with hotel owners • AirBnB's use of data analytics and data science to enhance collaboration among guests, hosts, and other ecosystem stakeholders
	Creating and managing ecosystem-based business models	• The hospitality industry has a growing and diverse number of ecosystems and organizations playing roles such as ecosystem orchestrators, coordinators, or creators (e.g., Airbnb, TripAdvisor, Open Table, Accor - through their MAX and WeMAX ecosystem)
	Enhancing organization adaptability, agility, and ambidexterity	• Some hospitality organizations are using big data and search engine data to anticipate and better respond to emerging events (e.g., Covid-19), changes in customer behaviors (e.g., changes in customer expectations and customer journeys), and changes in competitor activities

change process requires structural, cultural, capability, competency, and infrastructure changes aimed at delivering the necessary digitalized business models, business processes, products/services, and assets (e.g., see Perelygina et al., 2021; Pesonen, 2020; Vial, 2019; Hess et al., 2016; Matt et al., 2015). And effectively competing as a digital business requires H&T organizations to optimize digitalized business models, processes, products/services, and assets – in order to maximize competitive advantage and business longevity benefits on offer (e.g., see Alrawadieh et al., 2021; Farías and Cancino, 2021; Pesonen, 2020).

Information systems and management scholars have identified a range of new or digitally enhanced organization capabilities H&T organizations require to effectively undertake digital transformation, and to effectively compete as digital businesses (e.g., Warner and Wäger, 2019; Matt et al., 2015). When we refer to a "new" capability, we mean that capability constitutes the introduction of a completely new group of organization routines/activities. Whereas an "digitally enhanced" capability is one for which new digitally related activities are added to or augment existing routines/activities within the capability so as to significantly transform that capability. It is highly likely that hospitality managers need to play crucial roles in the building and sustenance of digital transformation and digital business-related capabilities. Therefore, identification and more specific understanding of these capabilities and the associated managerial roles related to them may enable us to infer related managerial digital transformation and digital business-related competencies. In turn, inference of these competencies may enable us to resolve the disconnect between the hospitality management DT competencies identified in the hospitality management literature to date, the organization capabilities required for digital transformation and digital business competitiveness, and the management competencies required to build and sustain these capabilities.

3. Methodology

The conceptual research design for this study pragmatically combined and followed those of Amani and Fadlalla (2017), Becker and Jaakkola (2020), and Okumus (2003). Our design varied from each, where necessary, in line with the prescriptions of Jaakkola (2020), Khlif and Chalmers (2015), Lukka and Vinnari (2014), and Tranfield et al. (2003) relating to the optimal approaches in using theory and the benefits of pragmatically combining different conceptual approaches. Drawing on these designs and prescriptions, our study's design was made up of the following steps:

3.1. Step 1. Identification of domain theory and starting point

As per the prescriptions of Jaakkola (2020), we identified the domain theory and starting point of the study. The relevant domain theory is the H&T literature identifying required managerial DT competencies, as well as the literature organizing such competencies into categories or a framework to enhance overall understanding. Within this domain theory, we started with the most recent literature organizing DT competencies into categories or frameworks – Bilgihan et al. (2014). Thus, the aim of this study is to make core contributions to both the literature identifying required competencies, as well as the integrative framework literature that seeks to enhance understanding of overall required managerial DT competencies and their interrelationships.

3.2. Step 2. Identification of method theory and uses of method theory

We followed the guidance of Jaakkola (2020) and Lukka and Vinnari (2014) to identify the method theory and its uses. The method theory for this study, or literature to be used as data, is made up of three literature streams from three disciplines. The first literature stream is the H&T literature identifying required managerial DT competencies. The second is the H&T literature organizing required managerial DT competencies into categories or integrative frameworks. The third literature stream is the information systems and management literature discussing the implications of digital disruption and digital transformation for required organization capabilities. The first and third literature streams enabled illumination of the full composition of required managerial DT competencies, including digital transformation and digital business-related ones not yet discussed in the H&T literature. The second literature stream enabled identification of previously discussed interrelationships among competencies. And all three literature streams enabled refinement and extension of identified interrelationships within managerial DT competencies, as well with broader hospitality management competencies.

3.3. Step 3. Identification of search terms

To identify literature search terms, we followed the approaches of Amani and Fadlalla (2017) and Becker and Jaakkola (2020). Based on preliminary method theory review, key competency related, DT related, management/leadership related, organization capability related, and discipline related keywords were identified (see specific keywords in Table 3). Combinations of these keywords (e.g., competen* + digital* + manag* + hospitality) were used to surface relevant literature. We considered the different combinations of keywords possible from relevant keyword groups to be sufficient to surface most of the relevant literature.

Table 3
Keyword types and specific keywords combined in the literature search.

Keyword types or groups	Specific keywords
Competency related keywords	Competen*, skill*, ability*,
Leadership/management related keywords	Lead*, leader*, manag*, role*
Organization capability related keywords	Capabili*, strateg*, compet*, business
Digital technology related keywords	Digital*, disruption, transformation, technolog*, IT, ICT
Discipline related keywords	Hospitality, tourism, information systems

3.4. Step 4. Identification of data sources

Our data sources consisted of Google Scholar and university library e-resources (which comprise more than 360 databases, including most of the major ones like EBSCOhost Business Source Ultimate and Elsevier ScienceDirect). This choice of data sources was in line with the approaches of Amani and Fadlalla (2017) and Becker and Jaakkola (2020).

3.5. Step 5. Data collection

Collecting relevant articles involved searching for the keyword combinations discussed in step 3 within the two data sources identified in step 4. Then, a prospective article's synopsis was skim read for suitability. For expediency, once one journal article of sufficient quality identified a competency, a competency category, or a competency interrelationship, subsequent literature discussing that same concept was not collected. The justification for this was that subsequent search for better quality, more recent, or more specific corroborating literature would be undertaken as needed later in the data evaluation and synthesis stages. Regarding journal quality, an article's journal had to have a SCImago rank of Q2 or better to be used, and Q1 wherever possible. Our data collection approach was in line with those of Amani and Fadlalla (2017) and Becker and Jaakkola (2020). The main search was conducted between July 2019 and December 2020, although searches for literature to enhance corroboration and refinement continued iteratively until September 2021.

3.6. Step 6. Data filtering

Data filtering followed the approaches of Amani and Fadlalla (2017) and Becker and Jaakkola (2020). Data from step 5 were filtered for papers that met the following inclusion criteria: (1) papers discussing required managerial DT competencies (2) papers discussing managerial DT competency categories or integrative frameworks (3) papers discussing required digital transformation and digital business-related organization capabilities. As part of the filtering process, the article title and synopses were first reviewed during the search process. Then, for the resultant collection of articles, the full articles were skim read to ensure data filtering criteria compliance. The complying ones were included on the final list and their bibliographies were reviewed for additional compliant articles to extend the list.

3.7. Step 7. Data evaluation

Following Amani and Fadlalla's (2017) approach, a data extraction form was used during evaluation of each of the final articles to enable capturing of an article's:

- Bibliographic details (e.g., date, authors, title, journal).
- Identified managerial DT competencies.
- Identified managerial DT competency categories and interrelationships.
- Identified digital transformation and digital business-related organization capabilities and definitions.
- Discussed micro-capabilities (i.e., organization routines or activities) within each identified capability.
- Managerial digital transformation and digital business-related competencies identified or implied by new or enhanced DT related capabilities and related micro-capabilities.

The data extraction form was inductively developed during preliminary literature review and iteratively improved or refined throughout the literature search, filtering, and evaluation processes.

3.8. Step 8. Synthesis and conceptualization

We pragmatically drew on the synthesis and conceptualization approaches of Okumus (2003) and Becker and Jaakkola (2020). First, we synthesized the competency categories within existing frameworks as prescribed by Jaakkola (2020). This involved grouping competency categories across frameworks that essentially referred to the same competencies but with different names, those that overlapped or were in conflict, and those that were unique (e.g., see similar approach of Okumus, 2003). A new and more inclusive competency category was derived to accommodate each group of competencies - resulting in version 1 of the integrated framework (see Fig. 1). Second, we considered what competency categories and subcategories might be missing from this initial version of the framework. To identify them, we first attempted to fit each of the existing and new competencies into one of the version 1 framework categories. Whilst many competencies fit into a version 1 category, there remained a large list of competencies that did not neatly fit into any of the version 1 categories. Within this list of non-fitting competencies, we teased out similarities so as to organize them into subgroups from which we could deduce additional meaningful categories or subcategories. After deducing additional categories to fit all competencies into a category, we then more closely examined competencies within each category to tease out category names that better described the competencies within each category. This resulted in version 2 categories. Third, we organized the version 2 categories into groups of related categories and considered exactly what those groups of related categories had in common. This resulted in the identification of additional version 3 categories to complement version 2 categories, such that version 2 categories became subcategories of version 3 categories. We then considered if any competencies might be missing. It struck us that the breadth and depth of required DT competencies may be overwhelming and challenging to learn. We considered whether a unique technology learning competency was required. So, we revisited the literature for any discussion of technology learning challenges, strategies and or competencies. We came across branches of the information systems literature discussing technological knowledge renewal challenges and strategies for information technology professionals (e.g., see Rong and Grover, 2009). We also came across aspects of that literature discussing strategies for coping with information overload that were corroborated by H&T researchers (e.g., see Saxena and Lamest, 2018; Rong and Grover, 2009; Savolainen, 2007). Drawing on findings of these literatures, we added the *technology learning* competency category and associated competencies to version 2 categories.

Finally, we considered the potential interrelationships between DT competency categories and sub-categories, as well as their interrelationship with broader hospitality management competency categories. We were able to identify interrelationships within DT competency categories, and between DT competency categories and broader hospitality management competencies that were supported by the H&T literature or the information systems and management literature (these are depicted in Fig. 2). In lieu of approaches enabling quantitative measurement of inter coder reliability when synthesizing or deriving categories, we employed a gated negotiated agreement approach to accepting derivation of the new and more inclusive competency categories within our integrated framework (e.g., see O'Connor and Joffe, 2020 for acceptable alternative approaches to intercoder reliability). Specifically, a single researcher derived each new framework category and interrelationship but had to get it through the scrutiny of the second researcher, and then the third researcher, prior to acceptance for inclusion in the framework. The first researcher iteratively modified each derived category to accommodate issues raised by the other two researchers until agreement or acceptance was reached. Or until the first researcher eventually discarded the category or interrelationship. Several initial categories were discarded this way. Following this, researcher one had to fit all DT competencies into one or more of the framework categories. The overall competencies category fit had to again be accepted by the other two researchers or the categories tweaked until it was accepted (see Table 6 for final fit). This process of constructive challenge and iterative modifications to arrive at final agreement enhanced the inclusiveness, completeness, and meaningfulness of the final framework. The negotiated agreement approach was adapted from Campbell et al. (2013).

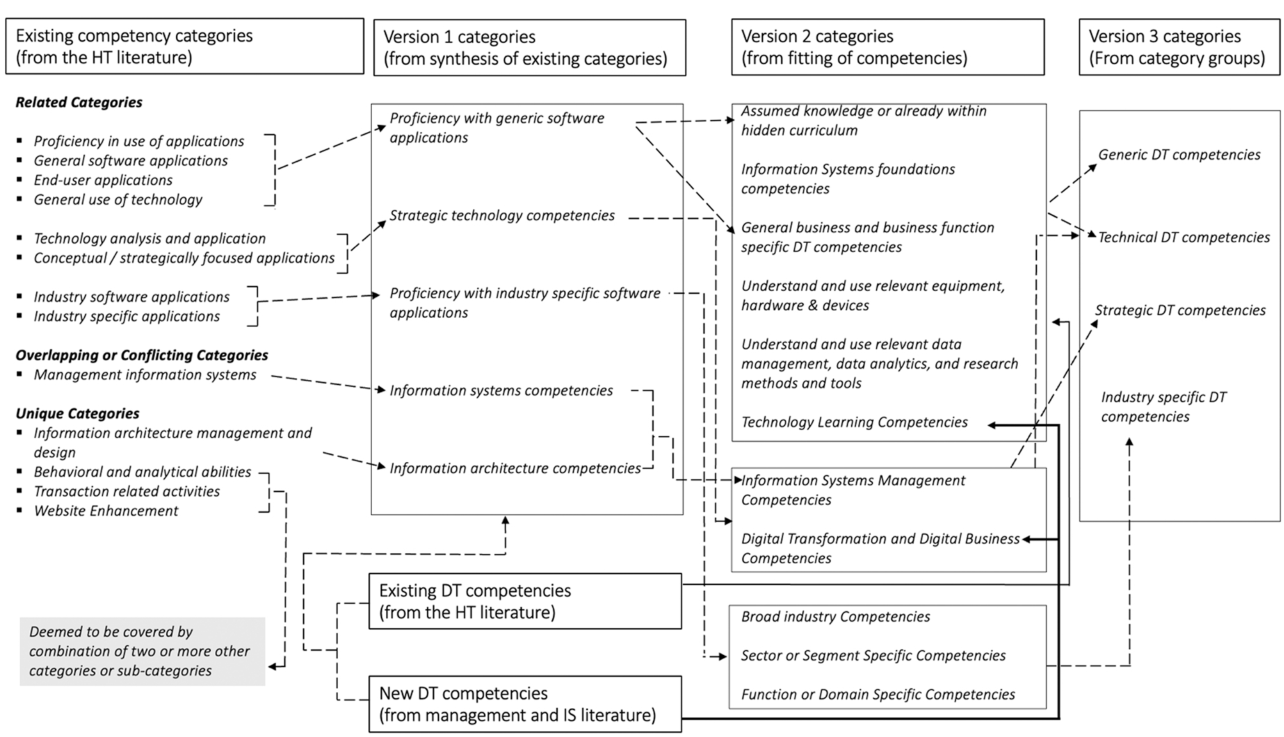

Fig. 1. Framework derivation from synthesis of existing competency categories, derivation of new categories, and identification of category interrelationships.

Table 4
Identified digital transformation and digital business capabilities, along with their definitions, and example micro-capabilities.

Digital transformation and digital business capability	Definition	Example micro-capabilities (groups of routines or activities)
Digital Transformation Strategy	Strategy that focuses on how to best undertake the change process involved in becoming a digital business (e.g., see Vial, 2019; Hess et al., 2016; Matt et al., 2015)	Adapted from Warner and Wäger (2019): • Building/leveraging internal enablers of digital transformation • Overcoming internal barriers to digital transformation • Redesigning internal structures to support digital transformation • Digital mindset crafting/effecting a digital culture
Digital Business Strategy	Refers to "…organizational strategy formulated and executed by leveraging digital resources to create differential value (e.g., Bharadwaj et al., 2013, p.472)	Adapted from Warner and Wäger (2019): • Digital scouting of opportunities and threats or strategies and practices • Digital scenario planning • Balancing digital portfolios • Building and optimizing digital assets and business models
Digital Innovation	Refers to "...the carrying out of new combinations of digital and physical components [in a layered modular architecture] to produce novel products". (Yoo et al., 2010, p.725) Digital innovation of business models, operating models, processes, and other forms of new value creation have been discussed in the literature (Warner and Wäger, 2019)	• Creation of digital/physical, programmable/reprogrammable, connected/smart products (Warner and Wäger, 2019) • Management/contribution to/use of distributed innovation, open innovation, network centric innovation, shared cognition, and joint sense-making approaches to innovation (e.g.,Yoo et al., 2010) • Management/contribution to/use of innovation collectives (Nambisan et al., 2017) • Management/contribution to/use of crowdsourcing, crowd funding, digital makerspace and other digital innovation platforms (Nambisan et al., 2017) • Leveraging rapid prototyping approaches (Warner and Wäger, 2019)
Data Analytics and Data Science	Data Analytics refers to ".the process of examining and analyzing raw data to obtain trends about the information they hold for decision-making in organizations" (Koohang and Nord, 2021, p.1) Data science is an interdisciplinary field that aims to turn data into real value. It can draw on approaches, methods, techniques, and theories from mathematics, statistics, computer science, and information science to provide value in the form of predictions, automated decisions, decision models learned from data, or visualizations of data insights (Van Der Aalst, 2016)	• Building and maintenance of data management capability (Aiken et al., 2007) • Optimizing use of and value from big data analytics (Mikalef et al., 2018; Gupta, & George, 2016) • Management of data science uses/applications (Kordon, 2020; Mazzei and Noble, 2017; Provost and Fawcett, 2013) • Management of data governance practice (Abraham et al., 2019)
Digital Ethics	Refers to the ethicality of the existing and potential impacts of DT uses on political, economic, social, environmental, and legal entities, dynamics, policies/rules, and expectations (e.g., see Mahieu et al., 2018; Floridi et al., 2019).	• Information privacy management (e.g., see Hajli et al., 2020; Bélanger and Crossler, 2011) • Fostering digital ethics culture (e.g., see Bélanger and Crossler, 2011; Vial, 2019) • Digital ethics governance (e.g., see Vial, 2019) • Digital ethics risk management (e.g., see Vial, 2019) • Digital ethics leadership (e.g., see Vial, 2019; Cortellazzo et al., 2019)
Digital Leadership or e-leadership	Refers to the "...social influence process mediated by Advanced Information Technology (AIT) to produce a change in attitudes, feelings, thinking, behavior, and/or performance with individuals, groups, and/or organizations" (Avolio et al., 2014, p. 617, cited in Cortellazzo et al., 2019, p.2).	Adapted from Cortellazzo et al. (2019): • Communicating effectively through digital media • Managing disruptive change • Managing connectivity • Ensuring leadership team proficiency with relevant digital tools • Leading virtual teams
Digital Customer Experience management	Refers to all customers' encounters with a company's products, services, and brands across digital channels/interfaces/devices – such as web, smartphones, tablets, platforms, apps (e.g., Holmlund et al., 2020; Zaki, 2019; Borowski, 2015)	• Creating and effecting digital customer experience strategy (e.g., Tivasuradej and Pham, 2019; Klaus, 2015) • Acquiring and using digital customer experience technologies (e. g., Holmlund et al., 2020; Zaki, 2019; Borowski, 2015 • Digital customer journey management (e.g., Holmlund et al., 2020; Zaki, 2019; Borowski, 2015) • Digital customer experience data management (e.g., see Holmlund et al., 2020) • Optimizing value from digital customer experience analytics (Holmlund et al., 2020)
Digital Customer Engagement	Refers to "...consumer's observable digital manifestations of brand engagement that go beyond purchase". (See Eigenraam et al., 2018, p.108)	Adapted from Eigenraam et al. (2018): • Leveraging digitally managed loyalty programs • Establishing and managing online brand communities • Leveraging digital customer engagement platforms • Leveraging customer co-creation practices • Leveraging consumer generated brand stories • Leveraging consumer engagement on social media • Leveraging consumer reviews
Digital Learning	Refers to learning that "...often happens spontaneously and unconsciously using technology without any prior stated objectives in terms of learning outcomes" - but it can also happen through leveraging technology and a structured process with specific learning materials, activities and evaluation methods" (Sousa and Rocha, 2018, p.564)	Adapted from Sousa and Rocha (2018): • Leveraging LMS/MOOCs/other learning platforms • Leveraging digital storytelling, technology integrated teaching, educational games, and other learning processes • Leveraging gamification, simulation, narrated stop motion animation, and other forms of learning facilitation

(*continued on next page*)

Table 4 (*continued*)

Digital transformation and digital business capability	Definition	Example micro-capabilities (groups of routines or activities)
		• Leveraging social, micro, and self-serve learning • Leveraging learning analytics • Leveraging personalized learning • Leveraging collaborative learning, cooperative learning, collaborative communities, networks and other forms of learning related participation
Workforce Digital Maturity or Workforce Digital Competencies or Workforce Digital Literacy	Refers to the nature and degree of DT competencies possessed or required by employees to enable effective use of DTs for the efficient and successful performance of organization work (e.g., see Oberländer et al., 2020; Warner and Wäger, 2019)	Adapted from Warner & Wäger (2019): • Identifying/measuring digital workforce maturity • External recruiting of digital talent • Developing the digital competencies of the existing workforce • Maximizing leveraging of digital knowledge inside the firm • Engaging and involving "digital natives" in capability building
Digital Culture	Refers to the set of shared assumptions, understanding of, and attitudes to organization functioning in a digital context – usually discussed in context of the extent to which they support or impede digital transformation (e.g., see Martínez-Caro et al., 2020).	Adapted from Warner and Wäger (2019): • Promoting a digital mindset • Cultivating an entrepreneurial mindset
DT Adoption and Use	Refers to the extent and effectiveness with which firms leverage DTs in their strategic and operational activities (e.g., Bharadwaj et al., 2013; Vial, 2019)	• DT sourcing (e.g., see Rothaermel and Alexandre, 2009) • Adoption and use of artificial intelligence for intelligent process automation (e.g., Bricher and Müller, 2020; Zhang, 2019) • Adoption and use of data analytics technologies for real time process monitoring (e.g., Nica et al., 2020) • Adoption and use of big data technologies for strategic sensing (Warner and Wäger, 2019)
Digital Governance	Refers to the practices and outcomes of establishing and implementing policies, procedures, and standards for the proper use of DTs and management of the unique issues/risks associated with them (e.g., see Almeida et al., 2020; Floridi et al., 2019).	• Establishing and maintaining effective guidelines for the use of DTs (e.g., Floridi et al., 2019) • Leveraging DTs to facilitate governance activities (e.g., see Vial, 2019)
Adaptability, Agility, and Ambidexterity	Adaptability refers to "the inbuilt capacity of an organization to change its strategies, structures, procedures or other core attributes, in anticipation or response to a change in its environment, including changes in relations with other organizations" (Hodgson et al., 2017, p.5) Agility refers to the "ability to detect opportunities for innovation and seize those competitive market opportunities by assembling requisite assets, knowledge, and relationships with speed and surprise" (Sambamurthy et al., 2003, p.245) Ambidexterity is defined as ".an organization's ability to be aligned and efficient in its management of today's business demands while simultaneously being adaptive to changes in the environment" (Raisch and Birkinshaw, 2008, p.375)	Adapted from Warner and Wäger (2019): • Scanning for technological trends • Sensing customer - centric trends • Analyzing market signals • Interpreting digital future scenarios • Rapidly reallocating resources • Pacing strategic responses • Speeding up decision making processes • Effecting faster and more flexible strategy making
Enterprise Architecture Management (EAM)	Refers to the establishment and continuous development of the design or configuration of the different elements or assets of an organization (e.g., hardware, software, networks, data, business processes) in order to optimize the organization's efficiency, effectiveness, and longevity in the pursuit of its mission (e.g., see Lange et al., 2016)	Adapted from Lange et al. (2016): • EAM organizational anchoring • Establishing or improving EAM services • Establishing or improving EAM infrastructure • Establishing or improving EAM products • Adapted from Kudryavtsev et al. (2018): • Designing Enterprise Architecture (EA) • Building EA stakeholder understanding and engagement • Participating in EA implementation • Participating in EA governance
Cybersecurity management	"Cybersecurity is the combination of people, policies, processes and technologies employed by an enterprise to protect its cyber assets" Gartner (n.d.-d.) or Cybersecurity refers to the state of or processes for protecting anything in the cyber realm and recovering from cyberattacks (Busulwa et al., 2020)	• Leading or supporting cybersecurity governance (e.g., Liu et al., 2020; Dutta and McCrohan, 2002) • Leading or supporting cyber risk management (e.g., see Paté-Cornell et al., 2018) • Adapted from NIST (2018): • Leading or supporting access control management • Leading or supporting recovery planning management • Leading or supporting cybersecurity risk mitigation • Leading or supporting security continuous monitoring

4. Findings

4.1. Digital transformation and digital business capabilities

The new or enhanced digital transformation and digital business capabilities identified in step 7 and step 8 (from evaluation of the management and information systems literature) are shown in Table 4, along with their definitions and examples of their associated micro-capabilities (i.e., groups of routines or activities). In relation to the bottom two capabilities, although cybersecurity management and enterprise architecture management are not new capabilities for most H&T organizations, digital transformation and digital business place greater demands on them. We therefore refer to them as digitally enhanced capabilities, since their practice needs to be more sophisticated to play critical support roles in the success of digital transformation and digital business efforts (e.g., see Vial, 2019; Warner and Wäger, 2019; Lange et al., 2016; Kudryavtsev et al., 2018). Although not specifically identifying them, the H&T literature has discussed aspects of some of the capabilities identified in Table 4. For example, the H&T literature has discussed aspects of capabilities such as digital transformation strategy (e.g., Lam and Law, 2019; Alrawadieh et al., 2021; Farías and Cancino, 2021), aspects of digital business strategy (e.g., Perelygina et al., 2021),

Table 5
Example of two digital transformation and digital business capabilities, their inferred managerial DT competencies, and the corresponding knowledge/skills/abilities relating to each competency.

Digital transformation or digital Business capability	Corresponding managerial DT competency	Example corresponding knowledge, skills, and abilities
Digital Innovation	Digital Innovation management	• Understand digital innovation concepts, processes, tools, aims • Apply knowledge of digital innovation concepts and practices to acquire or develop digital innovation workforce competencies (e.g. see Nylén and Holmström, 2015) • Understand, evaluate, and keep up to date with evolutions in digital devices, distribution channels, and user behaviors (Nylén and Holmström, 2015) • Understand, apply and evaluate distributed innovation, open innovation, network centric innovation, shared cognition, and joint sense-making approaches to innovation (e.g., Yoo et al., 2010) • Understand and apply innovation collective concepts and practices to manage/contribute to/use innovation collectives (e.g., see Nambisan et al., 2017) • Understand and apply crowdsourcing, crowdfunding, and digital makerspace concepts and practices so as to manage users or use related platforms (e.g., see Nambisan et al., 2017) • Analyze and or evaluate digital portfolios to be able to make or support digital portfolio configuration decisions (e.g., see Warner and Wäger, 2019) • Understand and apply knowledge of rapid prototyping concepts and practices to manage/facilitate/ support rapid prototyping practice. (e.g., see Warner and Wäger, 2019) • Create and or evaluate digital innovation strategies.
Digital Ethics	Digital Ethics management	• Understand digital ethics concepts, processes, tools, and aims. • Apply digital ethics concepts and practices to lead or contribute to the development and optimization of effective information privacy policies/procedures/practices (e.g., see Hajli et al., 2020; Bélanger and Crossler, 2011) • Apply effective strategies, attitudes, and behaviors to cultivate a digital ethics culture (e.g., see Bélanger and Crossler, 2011; Vial, 2019) • Apply digital ethics concepts and practices to ensure workforce compliance with digital ethics policies/ procedures/expectations • Analyze and evaluate governance of digital ethics risks (e.g., see Vial, 2019)

aspects of data analytics and data science (e.g., Stylos et al., 2021), aspects of digital leadership (e.g., Pesonen, 2020), aspects of DT adoption and use (e.g., Filimonau and Naumova, 2020; Gretzel et al., 2016; Ruel and Njoku, 2021), and aspects of adaptability/agility/ambidexterity (e.g., Stylos et al., 2021). We contend that an implicit assumption within the literature identifying these required capabilities and micro-capabilities is that hospitality managers are responsible for ensuring the building and sustenance of these capabilities. This is notwithstanding that the management responsibilities or roles may be shared between hospitality managers and functional or technical specialists (e.g., cybersecurity specialists, enterprise architects, digital strategists, IT managers).

4.2. Managerial digital transformation and digital business competencies

To the extent that hospitality managers have or share responsibility for the building and sustenance of the digital capabilities in Table 4, they require corresponding DT competencies to enable them to manage, lead, oversee, and or contribute to the development and sustenance of these capabilities. Thus, we propose the sixteen digital transformation and digital business capabilities in Table 4 translate into sixteen corresponding managerial digital transformation and digital business competencies (listed under "digital transformation and digital business competencies" in Table 6). We infer these competencies from the line management, leadership, oversight, or other workflow optimization roles that hospitality managers typically play in organization capabilities (e.g., see Sandwith, 1993; Mintzberg, 1990). Thus, in appending the terms "management" or "fostering" to a capability to create a competency name, we mean to conceptualize the knowledge, skills, and abilities relating to the roles of hospitality managers in that capability. So, for example, the managerial competency "digital ethics management" refers to the knowledge, skills, and abilities a hospitality manager requires in order to effectively manage, lead, oversee, or contribute to the ethical use of DTs by their direct reports and by employees in the wider organization.

In Table 5, we have used two example capabilities to show the logic of inferring competencies, along with some examples of the required knowledge, skills, and abilities relating to those competencies. That is, knowledge, skills, and abilities enabling hospitality managers to fulfill line management, leadership, oversight, or other contributary roles they may play in the building and sustenance of those two capabilities. The example provided in Table 5 can be adapted to each of the other digital transformation and digital business capabilities identified in Table 4. We framed the example knowledge, skills, and abilities in line with Krathwohl's (2002) revision of Bloom's taxonomy. It is true some capabilities are likely to be directly managed by a professional with relevant technical background (e.g., cybersecurity management capability and enterprise architecture management capability). However, even in situations where hospitality managers do not have direct management responsibility for such capabilities, they often still need to be involved in oversight, leadership, or contributary roles relating to that capability. For example, at strategic leadership level, a hospitality manager may be responsible for oversight of cybersecurity management capability. And at all management levels, he or she may be responsible for leading and managing compliance of direct reports with cybersecurity policies and procedures, as well as for contributing to cybersecurity related strategic initiatives. We propose that the sixteen DT competencies corresponding to the identified digital transformation and digital business capabilities add to and extend the existing hospitality management DT competencies list to reflect the need for hospitality managers to be able to enable or support the building and sustenance of their organizations' digital transformation and digital business capabilities.

4.3. Integrative framework

The final integrative framework, drawing on the expanded competency list, and derived as per step 8 of the methodology, is shown in Fig. 2. Consistent with the findings of Bilgihan et al. (2014), it shows that hospitality management DT competencies can be divided into technical DT competencies [1] and strategic DT competencies [2]. Technical DT competencies are the competencies required to be able to use DT tools, methods, and techniques to perform day to day operational management work. For example, a manager may need to know how to use food and beverage software, as well as spreadsheet software, in order to carry out performance monitoring and management of food and beverage personnel. Whereas Bilgihan et al. (2014) referred to these competencies

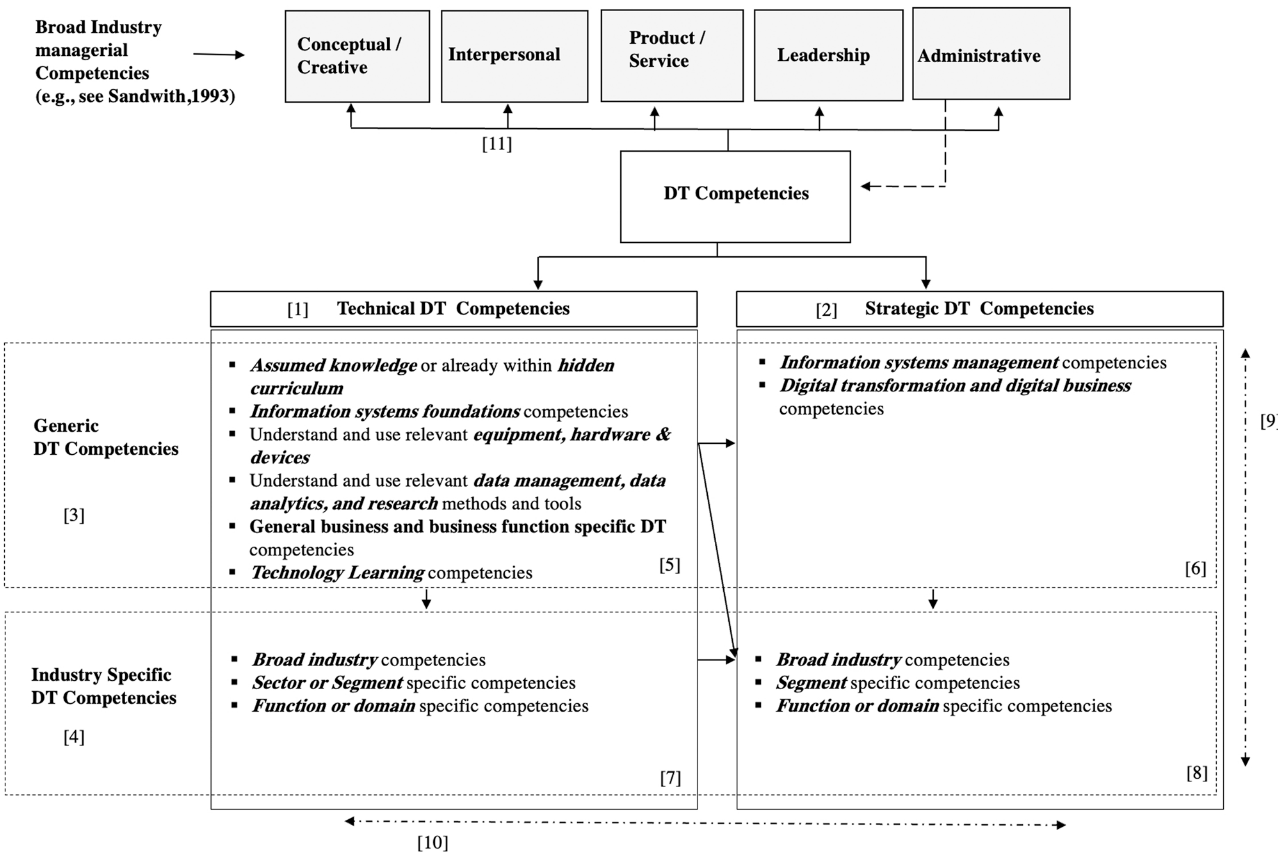

Fig. 2. Integrative hospitality management DT competency framework.

as "operational", we use the term "technical" since strategic competencies can also have a technical competency element that is not operational (e.g., understanding how a particular DT works can be a technical competency and also a strategic competency– as a prerequisite to understanding strategic implications of that DT).

In contrast to technical DT competencies, strategic DT competencies are the DT competencies required by managers to understand and manage strategic technology issues, as well as to be able to leverage DTs to change the position of an organization in its external environment or to enhance its longevity. For example, competencies like strategic information systems and digital innovation management are strategic DT competencies. Both technical and strategic DT competencies can further be divided into generic DT competencies [3] and industry specific DT competencies [4]. This is consistent with Bilgihan's (2014) proposal that the focus on IT applications ought to include generic and industry specific applications. Generic DT competencies are managerial DT competencies that are needed or used in almost all industries, and that are required by hospitality managers. For example, hospitality managers need to be proficient with the software applications that facilitate general managerial work (e.g., email applications like Microsoft Outlook) – a generic competency required or common to most managers across industries. And industry specific DT competencies are managerial DT competencies unique to the H&T industry (e.g., ability to use reservation systems like MARSHA and Amadeus). The juxtaposition of technical, strategic, generic, and industry specific DT competencies result in four unique competency quadrants (quadrants [5], [6], [7], and [8]). Line [9] depicts that competencies can exist on a continuum between quadrants [5] and [7] or between quadrants [6] and [8], such that some competencies may shift from one quadrant to another over time or have aspects that are in one quadrant as well as aspects that are in another quadrant. Similarly, line [10] depicts that competencies can be on continuum between quadrants [5] and [6] or quadrants [7] and [8]. Competencies may shift along continuums in line with technology, industry, or general business developments.

Within each quadrant are subgroups of competencies. Quadrant [5] contains subgroups of generic technical DT competencies such as information systems foundations competencies (e.g., as referenced by Su et al., 1997, and Dopson, 2005), and understanding and using relevant equipment/hardware/devices (adapted from Breiter and Hoart, 2000). The quadrant also includes general business and business function specific DT competencies (e.g., most of the ones discussed by Dopson, 2005, and Bilgihan et al., 2014), as well as understanding and using data management/data analytics/research methods and tools (adapted from Kay and Moncarz, 2004 and Cho and Connolly, 1999 cited in Sisson and Adams, 2013). Lastly, it includes technology learning competencies (mostly adapted from Rong and Grover, 2009). Since these technical competencies are generic, an aspect of them can be assumed – in line with the level of managerial education or practice being undertaken. For example, postgraduate hospitality management students could now be assumed to already be proficient in generic word processing, spreadsheet, and presentation software – whereas this may not have been the case when early industry DT competency research was undertaken. However, even if students are not already proficient in these applications, they are likely to implicitly become proficient through the necessity of using them to complete curriculum work – what Mandabach et al. (2001) referred to as the "hidden IT curriculum". These types of competencies are included under the subgroup "assumed knowledge or already within hidden curriculum".

Quadrant [7] contains industry specific technical DT competencies (these are adapted from the "industry specific applications" category proposed by Bilgihan et al. (2014)). These competencies may be relevant to most sectors and segments in the industry (as depicted by the subcategory "broad industry competencies"), only relevant to particular sectors and segments of the industry (as depicted by the "sector or

Table 6
How existing and new hospitality management DT competencies fit within the integrated DT competency framework.

	Technical DT Competencies	Strategic DT Competencies
Generic DT Competencies	***Assumed Knowledge or Already within Hidden Curriculum*** From Bilgihan et al. (2014): • Explain why a computer program is not working • Use email systems (i.e., Outlook Gmail) • Use word processing programs (e.g., Microsoft Word) • Use spreadsheet programs (e.g., Microsoft Excel) • Use presentation programs (e.g., Microsoft PowerPoint) • Use database programs (e.g., Microsoft Access) • Undertake desktop publishing • Delete viruses and spyware • Shop on the internet • Install computer software • Use web browsers (Breiter & Hoart, 2000) • Information literacy (Lowry & Flohr, 2005) ***Information Systems foundations competencies*** Knowledge and understanding of: – The purpose and design of information systems – Components of and types of information systems – Hardware types and components – Software types (e.g., operating and application software) – Network types (e.g., see Dopson, 2005) – Systems security and systems continuity (e.g., see Dopson, 2005) – EA foundations (e.g., see Dopson, 2005) – Emerging technology concepts and functioning (e.g., blockchain, edge computing) ***Understand and use relevant equipment, hardware & devices*** • Use communication equipment (Breiter and Hoart, 2000) • Use reproduction equipment (Breiter and Hoart, 2000) ***General business and business function specific DT competencies*** • General business DT competencies such as: – Create, maintain, or use different types of computer networks (e.g., Bilgihan et al., 2014) – Manage the use of intranets and extranets (Dopson, 2005) – Work effectively with website security and encryption (Dopson, 2005) – Use relevant workflow management/sharing/collaboration systems and platforms (e.g., Dopson, 2005) – Use project management software (e.g., Microsoft Project) (Bilgihan et al., 2014) – Use design and publishing software (e.g., Microsoft Publisher) (Bilgihan et al., 2014) – Use video recording and editing tools (Bilgihan et al., 2014) – Use graphic drawing programs (e.g., Photoshop) (Bilgihan et al., 2014) Business function specific DT competencies such as: – Develop and update websites (Dopson, 2005; Bilgihan et al., 2014) – Design web pages by using HTML, Java scripting, web development software, etc. (Dopson, 2005) – Use social networking tools (e.g., blogs, wikis, Facebook) (Bilgihan et al., 2014) – Use online marketing tools (Bilgihan et al., 2014) – Develop procedures and mechanics for launching a new business on the Internet (Dopson, 2005) – Manage information technology to build markets and customer relationships (Dopson, 2005) – Analyze e-marketing methods and implications (Dopson, 2005) – Have knowledge in e-commerce (Kay and Moncarz, 2004) – Use basic accounting software (Breiter and Hoart, 2000) – Use payroll software (Breiter and Hoart, 2000) ***Understand and use relevant data management, data analytics, and research related methods and tools*** • Analyze numeric data with computers (Bilgihan et al., 2014)	***Information Systems Management Competencies*** • Understand and apply technology sourcing and vendor management concepts and best practices (e.g., see Dopson, 2005; Rothaermel and Alexandre, 2009) • Understand and apply systems development lifecycle concepts and practices (e.g., see Bilgihan et al., 2014; Dopson, 2005) • Understand, design, analyze and evaluate strategic information systems • Apply IT project management knowledge and best practices to work effectively in IT project teams (Dopson, 2005) ***Digital Transformation and Digital Business Competencies*** (For each of the following, a manager should understand related concepts and be able to apply, analyze, or evaluate practices) • Digital innovation management • Management of digital business transformation strategy • Management of digital business strategy • Management of digital governance • Digital customer experience management • Management of digital customer engagement • Fostering adaptability/agility/ambidexterity • Management of data analytics and data science use • Leading digitally and fostering digital leadership • Management of DT adoption • Enterprise architecture management • Cybersecurity management • Management of digital ethics • Fostering digital culture • Management of workforce digital learning • Management workforce digital competencies

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Table 6 (*continued*)

	Technical DT Competencies	Strategic DT Competencies
	• Use business intelligence software (e.g., Cognos, Mirus) (Bilgihan et al., 2014) • Present data in an efficient manner (Bilgihan et al., 2014) • Undertake general programming (Breiter and Hoart, 2000) • Use the internet for resources and research (Kay and Moncarz, 2004) • Report generation and analysis (Mandabach et al., 2001) • Apply research methods and evaluation tools (Lowry and Flohr, 2005) **Technology Learning** • Apply technology learning strategies, tools, and practices (Rong and Grover, 2009) • Apply strategies and practices for dealing with DT related overload and overwhelm (e.g., see Saxena and Lamest, 2018) • Cultivate effective technology learning beliefs, attitudes, traits (Rong and Grover, 2009)	
Industry Specific DT Competencies	***Broad industry Competencies*** • Understanding of and or ability to manage or use industry enterprise management systems and ERP systems or modules that are common across sectors (e.g., Oracle Cloud ERP for hospitality, Infor Hospitality Management Solution – see Bilgihan et al., 2014; Dopson, 2005; Kay and Moncarz, 2004) • Understanding of and or ability to manage or use systems/software, platforms/ecosystems, and hardware/devices that are both unique to H&T and widely used across sectors. For example: – hospitality cost control/inventory software (Bilgihan et al., 2014) – Hospitality scheduling software (Breiter and Hoart, 2000) – Hospitality purchasing software (Breiter and Hoart, 2000) – Hospitality e-purchasing systems (Bilgihan et al., 2014) – Hospitality e-distribution systems (e.g., to manage cost of distribution, travel agency commissions, GDS fees – see Dopson, 2005) – -Hospitality point of sale technology (e.g., Simphony POS, Flipdish) – Hospitality property management systems (e.g., OPERA Cloud PMS) – Hospitality revenue management systems (e.g., IDeaS and Infor EzRMS) – Hospitality analytics and reporting systems (e.g., Oracle Hospitality Reporting and Analytics) ***Sector or Segment Specific Competencies*** • Understanding of and ability to manage or use systems/software, platforms/ecosystems, and hardware/devices that are unique to specific sectors or segments (e.g., Rezdy/Travefy/Oracle Hospitality Cruise Fleet Management systems for travel and tourism, Hotelogix/MARSHA/Amadeus systems for lodging) ***Function or Domain Specific Competencies*** • Understanding of and ability to use systems/software, platforms/ecosystems, and hardware/devices that are unique to specific domains within the industry. For example: – Hotel front desk applications (Mandabach et al., 2001) – Nutrition analysis software (Bilgihan et al., 2014) – Recipe applications (Mandabach et al., 2001) – Food and beverage management systems – Casino management systems	***Broad industry Competencies*** • Understand and evaluate the evolving role/uses of DTs in the industry (e.g., Bilgihan et al., 2014) • Understand and evaluate the latest DT trends in the industry (e.g., Bilgihan et al., 2014) • Understand and evaluate industry specific implications of critical DT issues (e.g., ethics, privacy, cybersecurity) • Understand and apply evolving industry digital transformation/digital business capability best practices ***Sector or Segment Specific Competencies*** • Evaluate the role of DTs in the sector or segment (e.g., see Bilgihan et al., 2014) • Understand the latest DT trends in the sector or segment (e.g., see Bilgihan et al., 2014) • Keep up with digital transformation/digital business capability best practices within the sector or segment ***Function or Domain Specific Competencies*** • Understand the latest DT applications in the function/domain (e.g., see Bilgihan et al., 2014) • Understanding the implications of digital transformation and digital business capabilities for specific functions/domains (e.g., for personalization of guest services, for customer engagement, for loyalty) • Keep up with and apply the latest DTs to relevant functions/domains (e.g., applying 3D printing in food and beverage to enhance making of deserts, using artificial intelligence to optimize guest comfort, or using virtual/augmented reality for virtual tours of key destinations)

segment specific” sub-category), or only relevant to particular H&T organization functions or domains (as depicted by the “function or domain specific” sub-category). We use the terms “function” and “domain” to refer to specialist H&T organization departments/units or value creation areas. Examples of such functions or domains include food and beverage, guest services, rooms division, gaming, and revenue management. So, in relation to function or domain specific competencies, a food and beverage manager may need to be competent in nutrition analysis and recipe management software – which is unique to the food and beverage domain. Whereas a guest services manager may need to be competent in hotel front desk software – which is unique to that domain. Function or domain specific competencies are still relevant, although at times some may be of lower priority, to hospitality managers not directly managing those functions or domains. In contrast, sector or segment specific competencies are unlikely to be needed by a manager or prospective manager outside of the relevant sectors or segments. Quadrant [6] contains generic strategic DT competencies that include the subgroups of information systems management and digital transformation and digital business competencies. And quadrant [8] contains strategic DT competencies that are unique to the industry. These may be relevant to the whole industry, only be relevant to particular sectors/segments, or only relevant to particular functions/domains. The broad industry, sector or segment specific, and function or domain specific groups of competencies enable illumination and compartmentalization of unique

DT competencies required to manage the full diversity of H&T services or activities. Altogether, there are 22 different competency categories (Two columns of competencies, plus two rows of competencies, plus the 4 quadrants of competencies, plus the fourteen bullet points representing subgroups of competencies within each quadrant).

Line [11] conceptualizes the relationship between DT competencies and broader hospitality management competencies discussed in Sandwith's (1993) managerial domains framework. The lines pointing to each domain depict that DT competencies are a catalyst for both the cultivation and effective practice of broader hospitality management competencies. For example, DT competencies enhance the ability to learn and practice leadership competencies and interpersonal competencies (e.g., consider findings by Cortellazzo et al. 2019 relating to the need for digital leadership competencies). And since DTs are increasingly the settings and tools through which conceptual/creative, product/service delivery (we use this term to replace the term "technical" as per Sandwith's original model to avoid confusion), and administrative domain workflows are facilitated, then DT competencies are critical to enhancing competencies within these domains. The dotted line from Sandwith's (1993) "administrative" competencies depicts that DT competencies could be taken as a subset of Sandwith's administrative competencies.

Table 6 shows how existing and new DT competencies fit within the integrated DT competency framework. The makeup of DT competencies includes some competencies from digitization, digitalization, and digital transformation evolution imperatives. As a result, there are some competency overlaps. For example, deleting viruses and spyware overlaps with cybersecurity management competencies. We propose that these overlaps be acceptable, since organizations are at different points along the evolution continuum from digitization to digitalization to digital transformation. Thus, the competencies required of a hospitality manager to be effective in organizations at each of these different points can differ. For example, deleting viruses and spyware is a competency that may still be a required by a manager in an organization still at the digitization point, whereas a manager in a digitally mature organization will instead require cybersecurity management competencies.

5. Contributions and future research directions

We add to theory on hospitality management DT competencies and their interrelationships with each other, as well as with broader hospitality management competencies. We do this through three conceptual contributions. Our first contribution is in integrating related findings from three distinct research disciplines (H&T, information systems, and management) to expand the composition of required managerial DT competencies for the industry (i.e., by adding digital transformation/digital business related and DT learning related competencies). This expansion of DT competencies addresses the concerning disconnect we found between prior hospitality management DT competency requirements literature (e.g., Bilgihan et al., 2014; Dopson, 2005), and the literature on required digital transformation and digital business capabilities owing to digital disruption (e.g., Vial, 2019; Warner and Wäger, 2019; Bharadwaj et al., 2013; Yoo et al., 2010). Left unaddressed, this disconnect risked hindering the progress of a critical hospitality management DT competencies literature; and therefore, risked limiting the ability of the industry to effectively navigate digital disruption.

Our second contribution is in addressing the proliferation of required managerial DT competencies, which is approaching some 100 required competencies (see Table 6). Without sufficiently meaningful structure to guide interpretation and use, this proliferation of competencies also risked limiting progress of the literature. We address this issue by developing a broader and yet more granular framework consisting of sufficient categories and subcategories for all existing competencies as well as new competencies to fit (see Table 6 for competencies fit). In doing so, we offer more complete guidance by resolving issues of overlapping/conflicting or missing competency categories and subcategories within prior frameworks. We also implicitly point to areas that may need further research, which is important for enabling progress of the literature. For example, the framework makes it more apparent which DT competency categories or subcategories may need identification of additional competencies. It also makes it more apparent what categories or subcategories require more specific understanding (e.g., looking at Table 6, it appears there is a need for additional research to expand and provide more specific understanding of subgroups within general business and business function specific DT competencies. And once this is done, perhaps some or all competencies related to data and research may fit within that category - or it may be better for them to remain in their current category. There is also need for expansion and specificity in relation to the composition of sector or segment specific and function or domain specific competencies).

Third, we propose interrelationships within DT competencies, and between DT competencies and broader hospitality management competencies. For example, the framework shows generic technical competencies aiding the cultivation of all other DT competencies; and generic strategic DT competencies aiding the cultivation of broad industry, sector or segment, and domain or function specific strategic DT competencies. Understanding these interrelationships has important implications for sequencing and scaffolding learning activities. Whilst our propositions regarding these interrelationships can do with empirical testing, we nevertheless provide a starting point for this to occur.

The extension of DT competencies and integrative understanding offered by this study builds a path to several much-needed research directions within the hospitality management DT competencies literature. Future research is now better positioned to explore and discuss how the presence or absence, and strength of particular competency categories, subcategories, or individual competencies, impacts managerial performance in general, and in particular settings or career stages (e.g., see Kay and Russette, 2000). This is important to address as it would provide future researchers, educators, and practitioners with additional clarity on which competencies may be of priority in those settings and career stages. For example, Kay and Moncarz (2004) had proposed that IT competencies were more important to lower-level managers. But with the delineation between technical and strategic DT competencies, it may be that technical DT competencies are more important to lower-level managers, whereas strategic DT competencies are more important to higher level managers.

Future research could explore whether particular competencies are growing or declining in relevance both broadly and within specific competency categories, subcategories, sectors or segments, and functions or domains. This would provide guidance to educators and practitioners regarding which competencies to divest from in order to invest in new ones. Future researchers could explore the minimum viable competency level for competency categories, subcategories, and individual competencies across career levels, across hospitality management roles, across sectors or segments, and across functions or domains. This would provide guidance to educators and practitioners regarding optimal learning benchmarks for different DT competencies at different career points. While we have identified specific digital transformation and digital business competencies, future research could unpack the specific and full set of knowledge, skills, and abilities that hospitality managers require within each of these competencies. Finally, the list of DT competencies remains too large, future research could explore duplications and overlaps in specific competencies (as we have done with competency categories). Whilst such a task would be a difficult undertaking with close to 100 competencies, researchers can use our DT competency framework to carve off subgroups of competencies for more manageable analysis and synthesis.

6. Framework implications for curricula design and managerial practice

Our proposed managerial DT competency framework can benefit curricula design in several ways. First, it clarifies required digital transformation and digital business competencies. This enables curricula designers to consider which of these competencies are already reflected or still need reflection in curricula. This is critical to ensuring readiness of curricula for digital business transformation. Second, the framework provides an integrated view of required DT competencies by bringing them together into a single framework. In doing so, it provides a flexible lens for understanding the fit and significance of any DT competencies identified in future. By clarifying which DT competencies can be considered assumed knowledge, or already implicit in hidden curricula, the framework can help with decision making regarding which competencies need to be deliberately designed into curricula and which do not. By delineating between technical and strategic competencies, the framework provides guidance on which courses or units can have particular DT competencies designed into them (e.g., it implies that strategic competencies ought to be designed into postgraduate courses or units, and they can be included in information systems management courses or strategic management courses). The framework also makes it easier to determine competency prerequisites and sequencing of competency related concepts within courses (e.g., before covering industry specific strategic DT competencies, students may need to understand generic strategic management concepts and generic strategic DT concepts).

By delineating between generic, industry specific, sector or segment, and function or domain specific competencies, the framework enables educators to exercise flexibility in choosing curriculum delivery staff (e.g., generic competencies can be collated into courses able to be delivered by instructors who do not have industry knowledge). Finally, by classifying all new and existing competencies into 22 different groups, the framework enables curricula designers to better undertake targeted cultivation or accentuation of particular groups of competencies within a degree or program (e.g., a degree/program producing graduates for a particular industry segment may focus on accentuating the DT competencies specific to that segment, whereas one producing general managers or industry consultants may accentuate strategic competencies). Taken together, these benefits offer curricula designers an expanded, integrated, and more specific understanding of the DT competencies to incorporate in curricula, how and where to incorporate them, and how they can be leveraged to enhance development of broader hospitality managerial competencies.

For managerial practice, the framework may enable assessment of the distribution of DT competencies across management teams at all levels (e.g., it could be used to check if there are sufficient digital transformation and digital business DT competencies at all levels). Absence of particular competencies or of sufficient depth in particular competencies may constrain management team performance. The framework may also be useful in recruitment and selection processes (e.g., the framework could potentially guide assessment of the type, breadth, and depth of managerial DT competencies candidates possess). Finally, performance development staff may be able to use the framework to guide targeted development of managerial DT competencies (e.g., the framework could be used to assess a manager's proficiency across all DT competencies, and weak areas can be the focus of targeted competency development efforts). Individual managers could also potentially use the framework for targeted development of their own competencies (e.g., a manager may look at the framework and see that they have limited understanding of digital leadership, thus motivating them to target their self-directed learning efforts at that competency). Overall, the framework can be invaluable to practitioners for DT competency assessment, DT competency recruitment, and DT competency development.

7. Conclusions

This paper highlights the implications of digital disruption, digital transformation, and effectively competing as a digital business for H&T organizations and for hospitality management DT competencies. The paper then identifies specific DT competencies required by hospitality managers to support the building and sustenance of digital transformation and digital business capabilities within H&T organizations. Building on this, the paper proposes an integrated competency framework that classifies existing and new competencies into generic vs. industry specific, and technical vs. strategic competencies. Within these classifications, a range of subclassifications are also derived. Interrelationships within DT competency categories and between DT competencies and broader hospitality management competencies are identified. The framework offers a range of avenues for future research and can be a valuable guide for curricula design as well as managerial recruitment and development efforts.

Declarations of interest

Nil.

Data availability

No data was used for the research described in the article.

References

Abraham, R., Schneider, J., Vom Brocke, J., 2019. Data governance: a conceptual framework, structured review, and research agenda. Int. J. Inf. Manag. 49, 424–438.

Aiken, P., Allen, M.D., Parker, B., Mattia, A., 2007. Measuring data management practice maturity: a community's self-assessment. Computer 40 (4), 42–50.

Almeida, V., Filgueiras, F., Gaetani, F., 2020. Digital governance and the tragedy of the commons. IEEE Internet Comput. 24 (4), 41–46.

Alrawadieh, Z., Alrawadieh, Z., Cetin, G., 2021. Digital transformation and revenue management: evidence from the hotel industry. Tour. Econ. 27 (2), 328–345.

Amani, F.A., Fadlalla, A.M., 2017. Data mining applications in accounting: a review of the literature and organizing framework. Int. J. Account. Inf. Syst. 24, 32–58.

Avolio, B.J., Sosik, J.J., Kahai, S.S., Baker, B., 2014. E-leadership: re-examining transformations in leadership source and transmission. Leadersh. Q. 25 (1), 105–131.

Becker, L., Jaakkola, E., 2020. Customer experience: fundamental premises and implications for research. J. Acad. Mark. Sci. 48 (4), 630–648.

Bélanger, F., Crossler, R.E., 2011. Privacy in the digital age: a review of information privacy research in information systems. MIS Q. 35 (4), 1017–1041.

Bharadwaj, A., El Sawy, O.A., Pavlou, P.A., Venkatraman, N., 2013. Digital business strategy: toward a next generation of insights. MIS Q. 37 (2), 471–482.

Bilgihan, A., Berezina, K., Cobanoglu, C., Okumus, F., 2014. The information technology (IT) skills of hospitality school graduates as perceived by hospitality professionals. J. Teach. Travel Tour. 14 (4), 321–342.

Borowski, C., 2015. What a great digital customer experience actually looks like. Harvard Bus. Rev. 11 (9), 2–5.

Breiter, D., Hoart, H., 2000. Competencies in foodservice information technology expected by the foodservice industry of graduates of hospitality bachelor's degree programs in the United States. J. Hosp. Tour. Educ. 12 (2), 11–17.

Bricher, D., Müller, A., 2020. A Supervised machine learning approach for intelligent process automation in container logistics. J. Comput. Inf. Sci. Eng. 20 (3), 031006.

Buergermeister, J., 1983. Assessment of the educational skills and competencies needed by beginning hospitality managers. Hosp. Educ. Res. J. 8 (1), 38–53.

Buhalis, D., 2003. eTourism: Information Technology for Strategic Tourism Management. Pearson Education.

Buhalis, D., Harwood, T., Bogicevic, V., Viglia, G., Beldona, S., Hofacker, C., 2019. Technological disruptions in services: lessons from tourism and hospitality. J. Serv. Manag. 30 (4), 484–506.

Buhalis, D., Leung, R., 2018. Smart hospitality—interconnectivity and interoperability towards an ecosystem. Int. J. Hosp. Manag. 71, 41–50.

Buhalis, D., O'Connor, P., 2005. Information communication technology revolutionizing tourism. Tour. Recreat. Res. 30 (3), 7–16.

Busulwa, R., Evans, N., Oh, A., Kang, M., 2020. Hospitality Management and Digital Transformation: Balancing Efficiency, Agility and Guest Experience in the Era of Disruption. Routledge.

Campbell, J.L., Quincy, C., Osserman, J., Pedersen, O.K., 2013. Coding in-depth semistructured interviews: problems of unitization and intercoder reliability and agreement. Sociol. Methods Res. 42 (3), 294–320.

Cheng, M., Bosselman, R., 2016. An evaluation of the research chefs association's bachelor of science in Culinology® core competencies. J. Hosp. Tour. Educ. 28 (3), 127–141.

Cheung, C., Law, R., He, K., 2010. Essential hotel managerial competencies for graduate students. J. Hosp. Tour. Educ. 22 (4), 25–32.

Cho, W., Connolly, D.J., 1999. U.S.-based hospitality programs: looking to keep pace with today's high-tech industry needs. Int. J. Hosp. Inf. Technol. 1 (1), 27–42.

Christou, E., 2002. Revisiting competencies for hospitality management: contemporary views of the stakeholders. J. Hosp. Tour. Educ. 14 (1), 25–32.

Chung, K.Y., 2000. Hotel management curriculum reform based on required competencies of hotel employees and career success in the hotel industry. Tour. Manag. 21 (5), 473–487.
Chung-Herrera, B.G., Enz, C.A., Lankau, M.J., 2003. Grooming future hospitality leaders: a competencies model. Cornell Hotel Restaur. Adm. Q. 44 (3), 17–45.
Cortellazzo, L., Bruni, E., Zampieri, R., 2019. The role of leadership in a digitalized world: a review. Front. Psychol. 10, 1938.
Doorsamy, W., Paul, B.S., Marwala, T., 2020. The Disruptive Fourth Industrial Revolution. Springer International Publishing, Cham.
Dopson, L.R., 2005. Underlying dimensions of hotel e-commerce competencies and the predicted immediacy for their inclusion in hospitality curricula from the hospitality educator's perspective. J. Hosp.Tour. Educ. 17 (1), 22–32.
Dutta, A., McCrohan, K., 2002. Management's role in information security in a cyber economy. Calif. Manag. Rev. 45 (1), 67–87.
Eigenraam, A.W., Eelen, J., Van Lin, A., Verlegh, P.W., 2018. A consumer-based taxonomy of digital customer engagement practices. J. Interact. Mark. 44, 102–121.
Farías, A., Cancino, C.A., 2021. Digital transformation in the Chilean lodging sector: opportunities for sustainable businesses. Sustainability 13 (14), 8097.
Filimonau, V., Naumova, E., 2020. The blockchain technology and the scope of its application in hospitality operations. Int. J. Hosp. Manag. 87, 102383.
Floridi, L., Cath, C., Taddeo, M., 2019. Digital ethics: its nature and scope. In: Ohman, C., Watson, D. (Eds.), The 2018 Yearbook of the Digital Ethics Lab. Springer, Cham, pp. 9–17.
Gartner, n.d.-a. Digitization. In: Gartner Glossary. Retrieved December 11, 2020, from ⟨https://www.gartner.com/en/glossary⟩.
Gartner, n.d.-b. Digitalization. In: Gartner Glossary. Retrieved December 11, 2020, from ⟨https://www.gartner.com/en/glossary⟩.
Gartner, n.d.-c. Digital business transformation. In: Gartner Glossary. Retrieved December 11, 2020, from ⟨https://www.gartner.com/en/glossary⟩.
Gartner, n.d.-d. Cybersecurity. In: Gartner Glossary. Retrieved January 09, 2021, from ⟨https://www.gartner.com/en/glossary⟩.
Goodman Jr, R.J., Sprague, L.G., 1991. The future of hospitality education: meeting the industry's needs. Cornell Hotel Restaur. Adm. Q. 32 (2), 66–69.
Gretzel, U., Zhong, L., Koo, C., 2016. Application of smart tourism to cities. Int. J. Tour. Cities 2 (2), 216–233.
Gupta, M., George, J.F., 2016. Toward the development of a big data analytics capability. Inf. Manag. 53 (8), 1049–1064.
Gursoy, D., Swanger, N., 2004. An industry-driven model of hospitality curriculum for programs housed in accredited colleges of business. Journal of Hosp. Tour. Educ. 16 (4), 13–20.
Hajli, N., Shirazi, F., Tajvidi, M., Huda, N., 2020. Towards an understanding of privacy management architecture in big data: an experimental research. Br. J. Manag. 32 (2), 548–565.
Helfat, C.E., Winter, S.G., 2011. Untangling dynamic and operational capabilities: strategy for the (N) ever-changing world. Strat. Manag. J. 32 (11), 1243–1250.
Hess, T., Matt, C., Benlian, A., Wiesböck, F., 2016. Options for formulating a digital transformation strategy. MIS Q. Executive 15 (2), 123–139.
Hirschheim, R., 2008. Some guidelines for the critical reviewing of conceptual papers. J. Assoc. Inf. Syst. 9 (8), 432–441.
Hodgson, G., Herman, S., Dollimore, D., 2017. Adaptability and survival in small- and medium-sized firms. Ind. Corp. Change 17 (2), dtx039.
Holmlund, M., Van Vaerenbergh, Y., Ciuchita, R., Ravald, A., Sarantopoulos, P., Ordenes, F.V., Zaki, M., 2020. Customer experience management in the age of big data analytics: a strategic framework. J. Bus. Res. 116, 356–365.
Huang, Y.-L., Lin, C.-T., 2010. Management trainee core competencies in the hospitality industry: differences between managers and scholars. J. Hum. Resour. Hosp. Tour. 10 (1), 1–13.
Jaakkola, E., 2020. Designing conceptual articles: four approaches. AMS Rev. 10 (1–2), 18–26.
Johanson, M., Ghiselli, R., Shea, L.J., Roberts, C., 2011. Changing competencies of hospitality leaders: a 25-year review. J. Hosp. Tour. Educ. 23 (3), 43–47.
Kay, C., Moncarz, E., 2004. Knowledge, skills, and abilities for lodging management. Cornell Hotel Restaur. Adm. Q 45 (3), 285–298.
Kay, C., Russette, J., 2000. Hospitality-management competencies: identifying managers' essential skills. Cornell Hotel Restaur. Adm. Q. 41 (2), 52–63.
Khlif, H., Chalmers, K., 2015. A review of meta-analytic research in accounting. J. Account. Lit. 35, 1–27.
Klaus, P., 2015. CX strategies and management practices. In: Measuring Customer Experience. Palgrave Macmillan, London, pp. 22–29.
Koohang, A., Nord, J.H., 2021. Critical components of data analytics in organizations: a research model. Expert Syst. Appl. 166, 114118.
Kordon, A.K., 2020. How to operate Ai-based data science in a business. In: Applying Data Science. Springer, Cham, pp. 433–451.
Krathwohl, D.R., 2002. A revision of Bloom's taxonomy: an overview. Theory Pract. 41 (4), 212–218.
Kudryavtsev, D., Zaramenskikh, E., Arzumanyan, M., 2018. The simplified enterprise architecture management methodology for teaching purposes. In: Workshop on Enterprise and Organizational Modeling and Simulation. Springer, Cham, pp. 76–90.
Kuo, C.M., Chen, L.C., Tseng, C.Y., 2017. Investigating an innovative service with hospitality robots. Int. J. Contemp. Hosp. Manag. 29 (5), 1305–1321.
Lam, C., Law, R., 2019. Readiness of upscale and luxury-branded hotels for digital transformation. Int. J. Contemp. Hosp. Manag. 79, 60–69.
Lange, M., Mendling, J., Recker, J., 2016. An empirical analysis of the factors and measures of enterprise architecture management success. Eur. J. Inf. Syst. 25 (5), 411–431.
Lee, M.J., Huh, C., Jones, M.F., 2016. Investigating quality dimensions of hospitality higher education: from students' perspective. J. Hosp. Tour. Educ. 28 (2), 95–106.
Lee, M.J., Olds, D.A., Lee, C., 2010. Why students choose a hospitality and tourism program: a pilot study of US undergraduate students. J. Hosp. Tour. Educ. 22 (3), 20–26.
Liu, C.W., Huang, P., Lucas Jr, H.C., 2020. Centralized IT decision making and cybersecurity breaches: evidence from US higher education institutions. J. Manag. Inf. Syst. 37 (3), 758–787.
Lowry, L.L., Flohr, J.K., 2005. No student left behind: a longitudinal assessment of the competency-based framework used to facilitate learning in a capstone tourism course. J. Hosp. Tour. Educ. 17 (4), 28–35.
Lukka, K., Vinnari, E., 2014. Domain theory and method theory in management accounting research. Account. Aud. Account. J. 27 (8), 1308–1338.
Mahieu, R., van Eck, N.J., van Putten, D., van den Hoven, J., 2018. From dignity to security protocols: a scientometric analysis of digital ethics. Eth. Inf. Technol. 20 (3), 175–187.
Mandabach, K.H., VanLeeuwen, D., Bloomquist, P., 2001. Hospitality technology education: student successes in mastering the hidden curriculum. J. of Hosp. Tour. Educ. 13 (1), 49–56.
Martínez-Caro, E., Cegarra-Navarro, J.G., Alfonso-Ruiz, F.J., 2020. Digital technologies and firm performance: the role of digital organisational culture. Technol. Forecast. Social Change 154, 119962.
Matt, C., Hess, T., Benlian, A., 2015. Digital transformation strategies. Bus. Inf. Syst. Eng. 57 (5), 339–343.
Mazzei, M.J., Noble, D., 2017. Big data dreams: a framework for corporate strategy. Bus. Horiz. 60 (3), 405–414.
Meredith, J., 1993. Theory building through conceptual methods. Int. J. Operat. Prod. Manag. 13 (5), 3–11.
Mikalef, P., Pappas, I.O., Krogstie, J., Giannakos, M., 2018. Big data analytics capabilities: a systematic literature review and research agenda. Inf. Syst. e-Bus. Manag. 16 (3), 547–578.
Mintzberg, H., 1990. The manager's job: folklore and fact. Harvard Bus. Rev. 68 (2), 163–176.
Nambisan, S., Lyytinen, K., Majchrzak, A., Song, M., 2017. Digital innovation management: reinventing innovation management research in a digital world. MIS Q. 41 (1), 223–238.
Nelson, A.A., Dopson, L., 2001. Future of hotel education: required skills and knowledge for graduates of US hospitality programs beyond the year 2000-part one. J. Hosp. Tour. Educ. 13 (5), 58–67.
Nica, E., Janoškova, K., Kovacova, M., 2020. Smart connected sensors, industrial big data, and real-time process monitoring in cyber-physical system-based manufacturing. J. Self-Gov. Manag. Econ. 8 (4), 29–38.
NIST, 2018. Cybersecurity Framework Version 1.1. Retrieved 9th January 2021 from: ⟨https://www.nist.gov/cyberframework/framework⟩.
Nylén, D., Holmström, J., 2015. Digital innovation strategy: a framework for diagnosing and improving digital product and service innovation. Bus. Horiz. 58 (1), 57–67.
Oberländer, M., Beinicke, A., Bipp, T., 2020. Digital competencies: a review of the literature and applications in the workplace. Comput. Educ. 146, 103752.
O'Connor, C., Joffe, H., 2020. Intercoder reliability in qualitative research: debates and practical guidelines. Int. J. Qual. Methods 19, 1–13.
Okumus, F., 2003. A framework to implement strategies in organizations. Manag. Decis. 41 (9), 871–882.
O'Reilly III, C.A., Tushman, M.L., 2008. Ambidexterity as a dynamic capability: resolving the innovator's dilemma. Res. Organ. Behav. 28, 185–206.
Oskam, J., Boswijk, A., 2016. Airbnb: the future of networked hospitality businesses. J. Tour. Futures 2 (1), 22–42.
Ottenbacher, M., Harrington, R., Parsa, H.G., 2009. Defining the hospitality discipline: a discussion of pedagogical and research implications. J. Hosp. Tour. Res. 33 (3), 263–283.
Paté-Cornell, M.E., Kuypers, M., Smith, M., Keller, P., 2018. Cyber risk management for critical infrastructure: a risk analysis model and three case studies. Risk Anal. 38 (2), 226–241.
Perdue, J., Woods, R.H., Ninemeier, J., 2001. Club management competencies 2005: updated information for the classroom. J. Hosp. Tour. Educ. 13 (2), 20–33.
Perelygina, M., Kucukusta, D., Law, R., 2021. Digital business model configurations in the travel industry. Tour. Manag. 88 (2022), 104408.
Pesonen, J., 2020. Management and leadership for digital transformation in tourism. In: Xiang, Z., Fuchs, M., Gretzel, U., Höpken, W. (Eds.), Handbook of e-Tourism. Springer, Cham.
Poon, A., 1993. Tourism, Technology and Competitive Strategies. CAB International.
Predvoditeleva, M., Reshetnikova, K., Slevitch, L., 2019. Hospitality master's program competencies: the case of the Russian hotel industry. J. Hosp. Tour. Educ. 31 (4), 246–255.
Priyangika, K.D.T., Perera, B.A.K.S., Wickremanayake Karunaratne, T.L., 2020. Facilities management roles in the hotel industry: the skills and competencies required. J. Qual. Assur. Hosp. Tour. 21 (4), 454–473.
Provost, F., Fawcett, T., 2013. Data Science for Business: What you Need to Know about Data Mining and Data-analytic Thinking. O'Reilly Media, Inc.
Raisch, S., Birkinshaw, J., 2008. Organizational ambidexterity: antecedents, outcomes, and moderators. J. Manag. 34 (3), 375–409.
Raybould, M., Wilkins, H., 2005. Over qualified and under experienced: turning graduates into hospitality managers. Int. J. Contemp. Hosp. Manag. 17 (3), 203–216.
Ritter, T., Pedersen, C.L., 2020. Digitization capability and the digitalization of business models in business-to-business firms: past, present, and future. Ind. Mark. Manag. 86, 180–190.

Rong, G., Grover, V., 2009. Keeping up-to-date with information technology: testing a model of technological knowledge renewal effectiveness for IT professionals. Inf. Manag. 46 (7), 376–387.
Rothaermel, F.T., Alexandre, M.T., 2009. Ambidexterity in technology sourcing: the moderating role of absorptive capacity. Organ. Sci. 20 (4), 759–780.
Ruel, H., Njoku, E., 2021. AI redefining the hospitality industry. J. Tour. Futures 7 (1), 53–66.
Sambamurthy, V., Bharadwaj, A., Grover, V., 2003. Shaping agility through digital options: reconceptualizing the role of information technology in contemporary firms. MIS Q. 27 (2), 237–263.
Sandwith, P., 1993. A hierarchy of management training requirements: the competency domain model. Public Person. Manag. 22 (1), 43–62.
Savolainen, R., 2007. Filtering and withdrawing: strategies for coping with information overload in everyday contexts. J. Inf. Sci. 33 (5), 611–621.
Saxena, D., Lamest, M., 2018. Information overload and coping strategies in the big data context: evidence from the hospitality sector. J. Inf. Sci. 44 (3), 287–297.
Sebastian, I.M., Ross, J.W., Beath, C., Mocker, M., Moloney, K.G., Fonstad, N.O., 2017. How big old companies navigate digital transformation. MIS Q. Exec. 16 (3), 197–213.
Sisson, L.G., Adams, A.R., 2013. Essential hospitality management competencies: the importance of soft skills. J. Hosp. Tour. Educ. 25 (3), 131–145.
Slattery, P., 2002. Finding the hospitality industry. J. Hosp. Leisure Sport Tour. Educ. 1 (1), 19–28.
Sousa, M.J., Rocha, Á., 2018. Learning analytics measuring impacts on organisational performance. J. Grid Comput. 18 (3), 563–571.
Steinicke, F., 2016. Toward a virtual revolution. In: Being Really Virtual. Springer, Cham, pp. 33–44.
Stylos, N., Zwiegelaar, J., Buhalis, D., 2021. Big data empowered agility for dynamic, volatile, and time-sensitive service industries: the case of tourism sector. Int. J. Contemp. Hosp. Manag. 33 (3), 1015–1036.
Su, A.Y.L., Miller, J.L., Shanklin, C., 1997. An evaluation of accreditation curriculum standards for four-year undergraduate hospitality programs. J. Hosp. Tour. Educ. 9 (3), 75–79.
Tas, R.F., 1988. Teaching future managers. Cornell Hotel Restaur. Admin. Q. 29 (2), 41–43.
Tas, R.F., LaBrecque, S.V., Clayton, H.R., 1996. Property-management competencies for management trainees. Cornell Hotel Restaur. Admin. Q. 37 (4), 90–96.
Tesone, D.V., Ricci, P., 2005. Attributes of entry-level employees: hospitality and tourism managers seeking more than knowledge and skills. Journal of Appl. Manag. Entrep. 10 (2), 3–12.
Tivasuradej, Y.C.T., Pham, N., 2019. Advancing customer experience practice and strategy in Thailand. Asia Pac. J. Mark. Logist. 31 (2), 327–343.
Tranfield, D., Denyer, D., Smart, P., 2003. Towards a methodology for developing evidence-informed management knowledge by means of systematic review. Br. J. Manag. 14 (3), 207–222.
Umbreit, W.T., 1992. In search of hospitality curriculum relevance for the 1990s. Hosp. Tour. Educ. 5 (1), 71–74.
Van Der Aalst, W., 2016. Data science in action. In: Process Mining. Springer, Berlin/ Heidelberg, pp. 3–23.
Vial, G., 2019. Understanding digital transformation: a review and a research agenda. J. Strat. Inf. Syst. 28 (2), 118–144.
Warner, K.S., Wäger, M., 2019. Building dynamic capabilities for digital transformation: an ongoing process of strategic renewal. Long Range Plann. 52 (3), 326–349.
Watson, S., 2008. Where are we now? A review of management development issues in the hospitality and tourism sector: implications for talent management. Int. J. Contemp. Hosp. Manag. 20 (7), 758–780.
Weber, M.R., Crawford, A., Lee, J., Dennison, D., 2013. An exploratory analysis of soft skill competencies needed for the hospitality industry. J. Hum. Resour. Hosp. Tour. 12 (4), 313–332.
Weill, P., Woerner, S.L., 2013. Optimizing your digital business model. MIT Sloan Manag. Rev. 54 (3), 71–78.
Yang, L.-T., Partlow, C.G., Anand, J., Shukla, V., 2014. Assessing the competencies needed by hospitality management graduates in India. J. Hosp. Tour. Educ. 26 (4), 153–165.
Yoo, Y., Henfridsson, O., Lyytinen, K., 2010. Research commentary—the new organizing logic of digital innovation: an agenda for information systems research. Inf. Syst. Res. 21 (4), 724–735.
Zabel, D., 1997. Integrating the Internet into the hospitality curriculum: more than a listing of “cool” web sites. J. Hosp. Tour. Educ. 9 (4), 66–73.
Zaki, M., 2019. Digital transformation: harnessing digital technologies for the next generation of services. J. Serv. Mark. 33 (4), 429–435.
Zhang, C., 2019. Intelligent process automation in audit. J. Emerg. Technol. Account. 16 (2), 69–88.